

## Leander F. Thiele

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|                   |                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                         |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>EMPLOYMENT</b> | <b>Kavli IPMU, University of Tokyo</b><br>Project Assistant Professor (tenure-track) in Center for Data-Driven Discovery (CD3)                                                                                                                                                                                                                                                                                                      | 2024–present                                                                                                            |
| <b>EDUCATION</b>  | <b>Princeton University</b> , Physics, PhD<br>advisor: David N. Spergel<br>thesis: <i>Getting ready for new Data: Approaches to some Challenges in Cosmology</i>                                                                                                                                                                                                                                                                    | 2019–2024                                                                                                               |
|                   | <b>Perimeter Institute</b> , Theoretical Physics, MSc<br>advisor: Kendrick M. Smith, co-advisor: J. Colin Hill<br>thesis: <i>Capturing non-Gaussianity: Analytic model for the one-point probability distribution function of cosmological fields within the halo model</i>                                                                                                                                                         | 2018–2019                                                                                                               |
|                   | <b>University of Oxford</b> , Physics, BA First Class<br>ranked top of the cohort ( $\sim 130$ students) in years 2 and 3                                                                                                                                                                                                                                                                                                           | 2015–2018                                                                                                               |
| <b>HONORS</b>     | Kusaka Memorial Prize in Physics (Princeton)<br>Member of the German Academic Scholarship Foundation<br>Perimeter Scholars International Award (Perimeter)<br>Scott Prize for best performance in the 3rd year (Oxford)<br>Winton Capital Prize for best performance in the 2nd year (Oxford)<br>BP Scholarship (Oxford)<br>Rokos Award for summer research project (Oxford)                                                        | \$3000<br>€35k<br>CA\$45k<br>£400<br>£250<br>£2000<br>£800<br>2022<br>2015–2019<br>2018<br>2018<br>2017<br>2017<br>2016 |
| <b>FUNDING</b>    | FUTI Travel Award<br>JSPS KAKENHI 26K17136 (early-career), <i>Reconstruction of the Local Universe with Machine Learning</i><br>Royal Society International Collaboration Award Round 2 (Japan) 252140, with William Coulton, <i>Transformative joint analyses of transformative datasets</i><br>Google Research Ecosystem Grant for IPMU CD3<br>JSPS KAKENHI 24K22878 (research-activity startup), <i>AI staring into the void</i> | ¥350k<br>¥3.5M<br>£225k<br>¥11M<br>¥3M<br>2026<br>2026<br>2025<br>2025<br>2024                                          |
| <b>TEACHING</b>   | Astro-AI Asia Network (A3Net) summer school, Seoul, admin<br>Lecture for UTokyo GUC, <i>Scientific Programming</i><br>Lecture & Tutorial for ILANCE students, IPMU, <i>Introduction to Machine Learning</i><br>Astro-AI Asia Network (A3Net) summer school, Osaka, <i>Basic Deep Learning</i>                                                                                                                                       | 2025<br>2025<br>2025<br>2024                                                                                            |
| <b>SERVICE</b>    | reviewer for ApJ, MNRAS, PRD, JCAP, NeurIPS ML4PS, ICML ML4Astro                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                         |

## ADVISING

|                                                       |                  |           |
|-------------------------------------------------------|------------------|-----------|
| Alexis Tastet (Ecole Polytechnique Masters)           | visitor          | 2026      |
| Kristers Nagainis (Latvia/Leiden PhD)                 | assisted visitor | 2026-     |
| Di He (UTokyo PhD)                                    | assisted         | 2025-     |
| Lillie Szemraj (Princeton undergrad)                  | assisted visitor | 2025      |
| Felicia Xiao (MIT undergrad)                          | visitor          | 2025      |
| Baptiste Barthe-Gold (Ecole Polytechnique Masters)    | visitor          | 2025      |
| Jessica Cowell (Oxford DPhil → Manchester PD)         | assisted         | 2024-2025 |
| Akira Tokiwa (UTokyo PhD → industry)                  | assisted         | 2024-2025 |
| Cooper Jacobus (UC Berkeley undergrad → Stanford PhD) | visitor          | 2024      |
| Bonny Wang (CMU Masters → UChicago PhD)               | visitor          | 2024-     |

## PUBLICATIONS Citations: 1300, h-index: 20 (INSPIRE-HEP, as of 2026-04-11)

**L. Thiele**, *Bayesian Cosmic Void Finding with Graph Flows*, 2026,  
arXiv:2602.14630 [astro-ph.CO], OJAp 9

J. Armijo, **L. Thiele**, J. Liu, *Replicating weak-lensing summary-statistic covariances with normalizing flows*, 2026,  
arXiv:2601.20669 [astro-ph.CO]

G. Fabbian, F. Bianchini, A. Sabyr, J.C. Hill, C.C. Lovell, **L. Thiele**, D.N. Spergel, *A new constraint on the  $y$ -distortion with FIRAS: implications for feedback models in galaxy formation and cosmic shear measurements*, 2025,  
arXiv:2512.03038 [astro-ph.CO]

A. Tokiwa, A.E. Bayer, J. Armijo, J. Liu, R. Terasawa, **L. Thiele**, M. Alvarez, L. Blot, M. Takada, *Impact of Simulation Box Size for Weak Lensing: Replication and Super-Sample Effects*, 2025,  
arXiv:2511.20423 [astro-ph.CO]

B. Barthe-Gold, N.-M. Nguyen, **L. Thiele**, *Reconstructing the local density field with combined convolutional and point cloud architecture*, 2025,  
arXiv:2510.08573 [astro-ph.CO], poster at ML4PS workshop NeurIPS 2025

B.Y. Wang, **L. Thiele**, *Set-based Implicit Likelihood Inference of Galaxy Cluster Mass*, 2025,  
arXiv:2507.20378 [cs.LG], spotlight talk at ML4Astro workshop (colocated with ICML 2025)

J.A. Cowell, J. Armijo, **L. Thiele**, G.A. Marques, C.P. Novaes, D. Grandón, S. Cheng, M. Shirasaki, D. Alonso, J. Liu, *First Constraints from Marked Angular Power Spectra with Subaru Hyper Suprime-Cam Survey First-Year Data*, 2025,  
arXiv:2507.12315 [astro-ph.CO]

**L. Thiele**, A.E. Bayer, N. Takeishi, *Simulation-Efficient Cosmological Inference with Multi-Fidelity SBI*, 2025,  
arXiv:2507.00514 [astro-ph.CO], poster at ML4Astro workshop (colocated with ICML 2025)

E. Calabrese, J.C. Hill, H.T. Jense, A. LaPosta, et al (incl. **L. Thiele**), *The Atacama Cosmology Telescope: DR6 Constraints on Extended Cosmological Models*, 2025,  
JCAP 11, 063, arXiv:2503.14454 [astro-ph.CO]

C. Jacobus, **L. Thiele**, P. Harrington, J. Liu, Z. Lukic, *Enhancing Cosmological Simulations with Efficient and Interpretable Machine Learning in the Gabor Wavelet Basis*, 2024, poster at Workshop on Machine Learning and the Physical Sciences (NeurIPS 2024)

**L. Thiele**, *De-baryonifying halos via optimal transport*, 2024, [arXiv:2411.18399](#) [[astro-ph.CO](#)]

J. Armijo, G.A. Marques, C.P. Novaes, **L. Thiele**, J.A. Cowell, D. Grandón, M. Shirasaki, J. Liu, *Cosmological constraints using Minkowski functionals from the first year data of the Hyper Suprime-Cam*, 2025, MNRAS 537, 4, [arXiv:2410.00401](#) [[astro-ph.CO](#)]

C.P. Novaes, **L. Thiele**, J. Armijo, S. Cheng, J.A. Cowell, G.A. Marques, E.G.M. Ferreira, M. Shirasaki, K. Osato, J. Liu, *Cosmology from HSC Y1 Weak Lensing with Combined Higher-Order Statistics and Simulation-based Inference*, 2025, PRD 111, 8, [arXiv:2409.01301](#) [[astro-ph.CO](#)]

A. Kogut, E. Switzer, D. Fixsen, N. Aghanim, J. Chluba, D. Chuss, J. Delabrouille, C. Dvorkin, B. Hensley, J.C. Hill, B. Maffei, A. Pullen, A. Rotti, A. Sabyr, **L. Thiele**, E. Wollack, I. Zelko, *The Primordial Inflation Explorer (PIXIE): Mission Design and Science Goals*, 2025, JCAP 004, 020, [arXiv:2405.20403](#) [[astro-ph.CO](#)]

S. Cheng, G.A. Marques, D. Grandón, **L. Thiele**, M. Shirasaki, B. Ménard, J. Liu, *Cosmological constraints from weak lensing scattering transform using HSC Y1 data*, 2025, JCAP 01, 006, [arXiv:2404.16085](#) [[astro-ph.CO](#)]

D. Grandón, G.A. Marques, **L. Thiele**, S. Cheng, M. Shirasaki, J. Liu, *Impact of baryonic feedback on HSC Y1 weak lensing non-Gaussian statistics*, 2024, PRD 110, 103539, [arXiv:2403.03807](#) [[astro-ph.CO](#)]

G.A. Marques, J. Liu, M. Shirasaki, **L. Thiele**, D. Grandón, K.M. Huffenberger, S. Cheng, J. Harnois-Déraps, K. Osato, W.R. Coulton, *Cosmology from weak lensing peaks and minima with Subaru Hyper Suprime-Cam survey first-year data*, 2023, MNRAS 528, 3, [arXiv:2308.10866](#) [[astro-ph.CO](#)]

**L. Thiele**, E. Massara, A. Pisani, C. Hahn, D.N. Spergel, S. Ho, B. Wandelt, *Neutrino mass constraint from an Implicit Likelihood Analysis of BOSS voids*, 2024, ApJ 969, 89, [arXiv:2307.07555](#) [[astro-ph.CO](#)]

**L. Thiele**, G.A. Marques, J. Liu, M. Shirasaki, *Cosmological constraints from HSC Y1 lensing convergence PDF*, 2023, PRD 108, 123526, [arXiv:2304.05928](#) [[astro-ph.CO](#)]

A.M. Delgado, D. Anglés-Alcázar, **L. Thiele**, M. Ntampaka, S. Pandey, K. Lehman, R.S. Somerville, S. Genel, F. Villaescusa-Navarro, *Predicting the impact of feedback on matter clustering with machine learning in CAMELS*, 2023, MNRAS 526, 4, [arXiv:2301.02231](#) [[astro-ph.GA](#)]

D. Wadekar, **L. Thiele**, J.C. Hill, S. Pandey, F. Villaescusa-Navarro, D.N. Spergel, M. Cranmer, D. Nagai, D. Anglés-Alcázar, S. Ho, L. Hernquist, *The SZ flux-mass (Y-M) relation at low halo masses: improvements with symbolic regression and strong constraints on baryonic feedback*, 2022, MNRAS 522, 2, [arXiv:2209.02075](#) [[astro-ph.CO](#)]

B.K.K. Lee, W. Coulton, **L. Thiele**, S. Ho, *An exploration of the properties of cluster profiles for the thermal and kinetic Sunyaev-Zel'dovich effects*, 2022, MNRAS 517, 420, [arXiv:2205.01710](#) [astro-ph.CO]

**L. Thiele**, M. Cranmer, W. Coulton, S. Ho, D.N. Spergel, *Predicting the Thermal Sunyaev-Zel'dovich Field using Modular and Equivariant Set-Based Neural Networks*, 2022, MLST 3, 035002, [arXiv:2203.00026](#) [astro-ph.CO], poster at the Fourth Workshop on Machine Learning and the Physical Sciences (NeurIPS 2021)

**L. Thiele**, D. Wadekar, J.C. Hill, N. Battaglia, J. Chluba, F. Villaescusa-Navarro, L. Hernquist, M. Vogelsberger, D. Anglés-Alcázar, F. Marinacci, *Percent-level constraints on baryonic feedback with spectral distortion measurements*, 2022, PRD 105, 083505, [arXiv:2201.01663](#) [astro-ph.CO]

D. Wadekar, **L. Thiele**, F. Villaescusa-Navarro, J.C. Hill, D.N. Spergel, M. Cranmer, N. Battaglia, D. Anglés-Alcázar, L. Hernquist, S. Ho, *Augmenting astrophysical scaling relations with machine learning: application to reducing the SZ flux-mass scatter*, 2022, PNAS 120(12), [arXiv:2201.01305](#) [astro-ph.CO]

F. Villaescusa-Navarro, S. Genel, D. Anglés-Alcázar, L.A. Perez, P. Villanueva-Domingo, D. Wadekar, H. Shao, F.G. Mohammad, S. Hassan, E. Moser, E.T. Lau, L.F.M.P. Valle, A. Nicola, **L. Thiele**, Y. Jo, O.H.E. Philcox, B.D. Oppenheimer, M. Tillman, C. Hahn, N. Kaushal, A. Pisani, M. Gebhardt, A.M. Delgado, J. Caliendo, C. Kreisch, K.W.K. Wong, W.R. Coulton, M. Eickenberg, G. Parimbelli, Y. Ni, U.P. Steinwandel, V. La Torre, R. Dave, N. Battaglia, D. Nagai, D.N. Spergel, L. Hernquist, B. Burkhart, D. Narayanan, B. Wandelt, R.S. Somerville, G.L. Bryan, M. Viel, Y. Li, V. Irsic, K. Kraljic, M. Vogelsberger, *The CAMELS project: public data release*, 2022, ApJS 265, 54, [arXiv:2201.01300](#) [astro-ph.CO]

B. Maffei, M.H. Abitbol, N. Aghanim, J. Aumont, E. Battistelli, J. Chluba, X. Coulon, P. De Bernardis, M. Douspis, J. Grain, S. Gervasoni, J.C. Hill, A. Kogut, S. Masi, T. Matsumara, C. O Sullivan, L. Pagano, G. Pisano, M. Remazeilles, A. Ritacco, A. Rotti, V. Sauvage, G. Savini, S.L. Stever, A. Tartari, **L. Thiele**, N. Trappe, *BISOU: a balloon project to measure the CMB spectral distortions*, 2021, 16th Marcel Grossmann Meeting, [arXiv:2111.00246](#) [astro-ph.IM]

F. Villaescusa-Navarro, S. Genel, D. Anglés-Alcázar, **L. Thiele**, R. Dave, D. Narayanan, A. Nicola, Y. Li, P. Villanueva-Domingo, B. Wandelt, D.N. Spergel, R.S. Somerville, J.M. Zorrilla Matilla, F.G. Mohammad, S. Hassan, H. Shao, D. Wadekar, M. Eickenberg, K.W.K. Wong, G. Contardo, Y. Jo, E. Moser, E.T. Lau, L.F.M.P. Valle, L.A. Perez, D. Nagai, N. Battaglia, M. Vogelsberger, *The CAMELS Multifield Dataset: Learning the Universe's Fundamental Parameters with Artificial Intelligence*, 2021, ApJS 259, 61, [arXiv:2109.10915](#) [cs.LG]

F. Villaescusa-Navarro, S. Genel, D. Anglés-Alcázar, D.N. Spergel, Y. Li, B. Wandelt, **L. Thiele**, A. Nicola, J.M. Zorrilla Matilla, H. Shao, S. Hassan, D. Narayanan, R. Dave, M. Vogelsberger, *Robust marginalization of baryonic effects for cosmological inference at the field level*, 2021, [arXiv:2109.10360](#) [astro-ph.CO]

F. Villaescusa-Navarro, D. Anglés-Alcázar, S. Genel, D.N. Spergel, Y. Li, B. Wandelt, A. Nicola, **L. Thiele**, S. Hassan, J.M. Zorrilla Matilla, D. Narayanan, R. Dave, M. Vogelsberger, *Multifield Cosmology with Artificial Intelligence*, 2021, [arXiv:2109.09747](#) [astro-ph.CO]

- L. Thiele**, Y. Guan, J.C. Hill, A. Kosowsky, D.N. Spergel, *Can small-scale baryon inhomogeneities resolve the Hubble tension? An investigation with ACT DR4*, 2021, PRD 104, 063535, [arXiv:2105.03003](#) [astro-ph.CO]
- L. Thiele**, J.C. Hill, K.M. Smith, *Accurate Analytic Model for the Weak Lensing Convergence One-Point Probability Distribution Function and its Auto-Covariance*, 2020, PRD 102, 123545, [arXiv:2009.06547](#) [astro-ph.CO]
- L. Thiele**, F. Villaescusa-Navarro, D.N. Spergel, D. Nelson, A. Pillepich, *Teaching neural networks to generate Fast Sunyaev Zel'dovich Maps*, 2020, ApJ 902, 129, [arXiv:2007.07267](#) [astro-ph.CO]
- R. Cayuso, O.J.C. Dias, F. Gray, D. Kubizňák, A. Margalit, J.E. Santos, R.G. Souza, **L. Thiele**, *Massive vector fields in Kerr–Newman and Kerr–Sen black hole spacetimes*, 2020, JHEP 159, [arXiv:1912.08224](#) [hep-th]
- L. Thiele**, C.A.J. Duncan, D. Alonso, *Disentangling magnification in combined shear clustering analyses*, 2020, MNRAS 491, 1746, [arXiv:1907.13205](#) [astro-ph.CO]
- R. Cayuso, F. Gray, D. Kubizňák, A. Margalit, R.G. Souza, **L. Thiele**, *Principal Tensor Strikes Again: Separability of Vector Equations with Torsion*, 2019, PLB 795, 650, [arXiv:1906.10072](#) [hep-th]
- L. Thiele**, J.C. Hill, K.M. Smith, *Accurate analytic model for the thermal Sunyaev-Zel'dovich one-point probability distribution function*, 2019, PRD 99, 103511, [arXiv:1812.05584](#) [astro-ph.CO]
- F. Dinc, M. Medvidovic, **L. Thiele**, *Effective Geometry Monte Carlo: A Fast and Reliable Simulation Framework for Molecular Communication*, 2019, IEEE Access 7, 28635
- F. Dinc, **L. Thiele**, B. C. Akdeniz, *The effective geometry Monte Carlo algorithm: applications to molecular communication*, 2019, PLA 383, 2594, [arXiv:1809.06438](#) [cs.ET]

## TALKS

|                               |         |         |
|-------------------------------|---------|---------|
| Sydney PPC Seminar, remote    |         | 4/2026  |
| Fundamentals Seminar, IPMU    |         | 2/2026  |
| AI4HEP East Asia, KEK Tsukuba | invited | 1/2026  |
| AI4KMI, Nagoya                | invited | 1/2026  |
| 3rd Shanghai Assembly         |         | 11/2025 |
| 28th IBIS, Naha               | invited | 11/2025 |
| NPML conference, Tokyo        | invited | 10/2025 |
| ML workshop @ McGill          |         | 9/2025  |
| Mila, Montreal                | invited | 9/2025  |
| 21st Rencontres du Vietnam    | invited | 8/2025  |
| AI for Fugaku seminar         |         | 7/2025  |
| ML4Astro, UTokyo              |         | 5/2025  |
| LeCosPA Meets IPMU, Taipei    |         | 4/2025  |
| FOPM seminar, UTokyo          |         | 3/2025  |
| Voids @ CPPM, Marseille       | invited | 2/2025  |
| KICC, Cambridge UK            |         | 2/2025  |
| FAIRS-Japan workshop Nagoya   | invited | 12/2024 |
| 11th KIAS workshop Gyeongju   | invited | 10/2024 |

|                                          |         |         |
|------------------------------------------|---------|---------|
| Cosmo'24 Kyoto                           |         | 10/2024 |
| LSS Quest Osaka                          |         | 6/2024  |
| MPA cosmology seminar                    |         | 5/2024  |
| Yale cosmology seminar                   | invited | 2/2024  |
| AI4Phys @ IPMU                           |         | 1/2024  |
| CCA Cosmo x ML tristate meeting          |         | 10/2023 |
| CMB constellation meeting KIPAC Stanford |         | 10/2023 |
| DESI lunch Berkeley Lab                  |         | 10/2023 |
| BCCP seminar UC Berkeley                 |         | 10/2023 |
| IPMU CD3 seminar                         |         | 9/2023  |
| Institute d'Astrophysique Spatiale Orsay |         | 9/2023  |
| Cosmo'23 Madrid                          |         | 9/2023  |
| Nagoya                                   |         | 4/2023  |
| IPMU                                     |         | 3/2023  |
| Princeton gravity group                  |         | 2/2023  |
| UCL Physics & Astronomy                  |         | 9/2022  |
| IAS astro coffee                         |         | 3/2022  |
| AAS 239                                  |         | 1/2022  |
| Cosmology Talks                          |         | 1/2022  |
| Learn the Universe @ CCA                 |         | 8/2021  |
| CMB-S4 meeting                           |         | 8/2021  |
| MPA Garching                             |         | 10/2020 |
| German Astronomical Society              |         | 9/2020  |
| Perimeter Institute                      |         | 5/2020  |
| Princeton/IAS cosmo lunch                |         | 5/2020  |
| CCA Cosmo x ML                           |         | 5/2020  |